

WPA2 Client Connection Analysis Using Wireshark

1. Objective

The objective of this exercise is to understand the process followed by a wireless client while connecting to an access point configured with WPA2 security. The captured packets are analysed using Wireshark to study the 802.11 management frame exchange, security-related Information Elements, WPA2 4-way handshake and available radio information.

Capture Note: Due to limitations in capturing complete raw 802.11 frames using the available personal laptop and wireless adapter, a sample 802.11/WPA capture was used for packet-level learning and analysis.

2. WPA2 Client Connection Overview

The connection process begins with wireless network discovery, followed by 802.11 authentication and association. After association, the WPA2 4-way handshake establishes the encryption keys required for protected communication.

4. Packet Analysis

4.1 Probe Request

Purpose:

Client searches for the required wireless network.

No.	Time	Source	Destination	Protocol	Length	Info
701	44.207153			802.11	10	Acknowledgement, Flags=.....
702	44.237074	Siemens_41:bd:6e	Broadcast	802.11	110	Beacon frame, SN=433, FN=0, Flags=....., BI=100, SSID="martinet3"
703	44.314988	NokiaDanmark_3d:aa:...	Broadcast	802.11	54	Probe Request, SN=11, FN=0, Flags=....., SSID="martinet3"
704	44.339477	Siemens_41:bd:6e	Broadcast	802.11	110	Beacon frame, SN=434, FN=0, Flags=....., BI=100, SSID="martinet3"
705	44.347636	NokiaDanmark_3d:aa:...	Broadcast	802.11	54	Probe Request, SN=12, FN=0, Flags=....., SSID="martinet3"
706	44.348282	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....., BI=100, SSID="martinet3"
707	44.349370	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
708	44.350295	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
709	44.351210	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
710	44.352609	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
711	44.357027	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
712	44.363487	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	104	Probe Response, SN=435, FN=0, Flags=....R..., BI=100, SSID="martinet3"
713	44.441864	Siemens_41:bd:6e	Broadcast	802.11	110	Beacon frame, SN=436, FN=0, Flags=....., BI=100, SSID="martinet3"
714	44.544280	Siemens_41:bd:6e	Broadcast	802.11	110	Beacon frame, SN=437, FN=0, Flags=....., BI=100, SSID="martinet3"
715	44.545208	NokiaDanmark_3d:aa:...	Siemens_41:bd:6e	802.11	30	Authentication, SN=13, FN=0, Flags=.....
716	44.545432		NokiaDanmark_3d:aa:...	802.11	10	Acknowledgement, Flags=.....
717	44.546099	Siemens_41:bd:6e	NokiaDanmark_3d:aa:...	802.11	38	Authentication, SN=438, FN=0, Flags=.....

Frame 705: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)	0000	40 00 00 00 ff ff ff ff ff ff 00 16 bc 3d aa 57	@	W
IEEE 802.11 Probe Request, Flags=.....	0010	ff ff ff ff ff ff c0 00 00 09 6d 61 72 74 69 6e		martin
Type/Subtype: Probe Request (0x0004)	0020	65 74 33 01 08 82 84 8b 96 0c 12 18 24 03 01 0c	et3	\$...
Frame Control Field: 0x4000	0030	32 04 30 48 60 6c	2 0H'1	
Duration: 0 microseconds				
Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)				
Destination address: Broadcast (ff:ff:ff:ff:ff:ff)				
Transmitter address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)				
Source address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)				
BSS Id: Broadcast (ff:ff:ff:ff:ff:ff)				
Fragment number: 0				
Sequence number: 12				
[WLAN Flags:]				
IEEE 802.11 Wireless Management				

4.2 Probe Response

Purpose:

AP responds with network and capability information.

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▶ Frame 706: Packet, 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
▼ IEEE 802.11 Probe Response, Flags: .....
  Type/Subtype: Probe Response (0x0005)
  ▼ Frame Control Field: 0x5000
    .... ..00 = Version: 0
    .... 00.. = Type: Management frame (0)
    0101 .... = Subtype: 5
    ▶ Flags: 0x00
    .000 0001 0000 0010 = Duration: 258 microseconds
    ▶ Receiver address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Destination address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Transmitter address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Source address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    .... .... 0000 = Fragment number: 0
    0001 1011 0011 .... = Sequence number: 435
    [WLAN Flags: .....]
  ▼ IEEE 802.11 Wireless Management
    ▶ Fixed parameters (12 bytes)
    ▼ Tagged parameters (68 bytes)
      ▼ Tag: SSID parameter set: "martinet3"
        Tag Number: SSID parameter set (0)
        Tag length: 9
        SSID: "martinet3"
      ▶ Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), 18, 24, 36, 54, [Mbit/sec]
      ▶ Tag: DS Parameter set: Current Channel: 11
      ▶ Tag: ERP Information
      ▶ Tag: ERP Information
      ▶ Tag: Extended Supported Rates 6, 9, 12, 48, [Mbit/sec]
      ▶ Tag: Vendor Specific: Broadcom
      ▶ Tag: Vendor Specific: Microsoft Corp.: WPA Information Element

```

4.3 Information Elements and RSN

Purpose:

To examine the network capabilities and WPA2 security information advertised by the AP.

Look for:

- SSID
- Supported Rates
- Channel-related information
- RSN Information

4.4 Authentication

Important observation:

This management-frame authentication should not be confused with the WPA2 4-way handshake.

```

▶ Frame 715: Packet, 30 bytes on wire (240 bits), 30 bytes captured (240 bits)
▼ IEEE 802.11 Authentication, Flags: .....
  Type/Subtype: Authentication (0x000b)
  ▼ Frame Control Field: 0xb000
    .... ..00 = Version: 0
    .... 00.. = Type: Management frame (0)
    1011 .... = Subtype: 11
    ▶ Flags: 0x00
    .000 0001 0000 0010 = Duration: 258 microseconds
    ▶ Receiver address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Destination address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Transmitter address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Source address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    .... .... 0000 = Fragment number: 0
    0000 0000 1101 .... = Sequence number: 13
    [WLAN Flags: .....]
  ▼ IEEE 802.11 Wireless Management
    ▶ Fixed parameters (6 bytes)

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4.5 Association Request

The client requests to join the AP and advertises its capabilities.

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▶ Frame 719: Packet, 79 bytes on wire (632 bits), 79 bytes captured (632 bits)
▼ IEEE 802.11 Association Request, Flags: .....
  Type/Subtype: Association Request (0x0000)
  ▼ Frame Control Field: 0x0000
    .... ..00 = Version: 0
    .... 00.. = Type: Management frame (0)
    0000 .... = Subtype: 0
    ▶ Flags: 0x00
    .000 0001 0000 0010 = Duration: 258 microseconds
    ▶ Receiver address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Destination address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Transmitter address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Source address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    .... .... 0000 = Fragment number: 0
    0000 0000 1110 .... = Sequence number: 14
    [WLAN Flags: .....]
  ▼ IEEE 802.11 Wireless Management
    ▶ Fixed parameters (4 bytes)
    ▼ Tagged parameters (51 bytes)
      ▼ Tag: SSID parameter set: "martinet3"
        Tag Number: SSID parameter set (0)
        Tag length: 9
        SSID: "martinet3"
      ▶ Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), 18, 24, 36, 54, [Mbit/sec]
      ▶ Tag: Extended Supported Rates 6, 9, 12, 48, [Mbit/sec]
      ▶ Tag: Vendor Specific: Microsoft Corp.: WPA Information Element

```

4.6 Association Response

The AP accepts or rejects the association request.

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▶ Frame 722: Packet, 10 bytes on wire (80 bits), 10 bytes captured (80 bits)
▼ IEEE 802.11 Acknowledgement, Flags: .....
    Type/Subtype: Acknowledgement (0x001d)
    ▼ Frame Control Field: 0xd400
        .... ..00 = Version: 0
        .... 01.. = Type: Control frame (1)
        1101 .... = Subtype: 13
        ▶ Flags: 0x00
        .000 0000 0000 0000 = Duration: 0 microseconds
    ▶ Receiver address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    [WLAN Flags: .....]

```

5. WPA2 4-Way Handshake

Message 1 — AP to Client

Observation:

AP initiates the WPA2 key establishment process and sends its nonce-related information.

[illegible]

Message 2 — Client to AP

Observation:

Client responds with its nonce-related information and key confirmation data.

```
▶ Frame 728: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits)
▼ IEEE 802.11 Data, Flags: .....T
  Type/Subtype: Data (0x0020)
  ▼ Frame Control Field: 0x0801
    .... ..00 = Version: 0
    .... 10.. = Type: Data frame (2)
    0000 .... = Subtype: 0
    ▶ Flags: 0x01
      .000 0000 0010 1100 = Duration: 44 microseconds
    ▶ Receiver address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Transmitter address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Destination address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Source address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ STA address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    .... .... 0000 = Fragment number: 0
    0000 0000 1111 .... = Sequence number: 15
    [WLAN Flags: .....T]
  ▶ Logical-Link Control
  ▼ 802.1X Authentication
    Version: 802.1X-2001 (1)
    Type: Key (3)
    Length: 119
    Key Descriptor Type: EAPOL WPA Key (254)
    [Message number: 2]
    ▶ Key Information: 0x0109
      Key Length: 0
      Replay Counter: 0
      WPA Key Nonce: 233a5f1cb602abb9b327b58bd38a2bc0782d9c90bf426c102cb82308c2ca821a
      Key IV: 00000000000000000000000000000000
      WPA Key RSC: 0000000000000000
      WPA Key ID: 0000000000000000
      WPA Key MIC: 628bb961ecec0e04fe28379e78c20f4a
      WPA Key Data Length: 24
    ▶ WPA Key Data: dd160050f20101000050f20201000050f20201000050f202
```

Message 3 — AP to Client

Observation:

AP sends the information required for the next stage of key installation and group-key configuration.

```
▶ Frame 733: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits)
▼ IEEE 802.11 Data, Flags: .....F.
  Type/Subtype: Data (0x0020)
  ▼ Frame Control Field: 0x0802
    .... ..00 = Version: 0
    .... 10.. = Type: Data frame (2)
    0000 .... = Subtype: 0
    ▶ Flags: 0x02
    .000 0000 0010 1100 = Duration: 44 microseconds
    ▶ Receiver address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Transmitter address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Destination address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Source address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ STA address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    .... .... 0000 = Fragment number: 0
    0001 1011 1001 .... = Sequence number: 441
    [WLAN Flags: .....F.]
  ▶ Logical-Link Control
  ▼ 802.1X Authentication
    Version: 802.1X-2001 (1)
    Type: Key (3)
    Length: 119
    Key Descriptor Type: EAPOL WPA Key (254)
    [Message number: 3]
    ▶ Key Information: 0x01c9
      Key Length: 32
      Replay Counter: 1
      WPA Key Nonce: f19c5c8014f2fdf875abfc0ccc238ae0272b97110c1e91d0b07e5fbbfc8722a9
      Key IV: 00000000000000000000000000000000
      WPA Key RSC: 0000000000000000
      WPA Key ID: 0000000000000000
      WPA Key MIC: 2aedcba325bdf4fcb279402c612b395b
      WPA Key Data Length: 24
    ▶ WPA Key Data: dd160050f20101000050f20201000050f20201000050f202
```

Message 4 — Client to AP

Observation:

Client confirms successful key installation, completing the 4-way handshake.

[illegible]

6. Protected Data

After the successful WPA2 key establishment process, data communication takes place using protected wireless frames. The payload is encrypted using the negotiated security keys and cannot be directly interpreted without the appropriate decryption information.

```
▶ Frame 743: Packet, 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
▼ IEEE 802.11 Data, Flags: .p....F.
  Type/Subtype: Data (0x0020)
  ▶ Frame Control Field: 0x0842
    .... ..00 = Version: 0
    .... 10.. = Type: Data frame (2)
    0000 .... = Subtype: 0
    ▶ Flags: 0x42
      .000 0000 0010 1100 = Duration: 44 microseconds
    ▶ Receiver address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Transmitter address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ Destination address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    ▶ Source address: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ BSS Id: Siemens_41:bd:6e (00:01:e3:41:bd:6e)
    ▶ STA address: NokiaDanmark_3d:aa:57 (00:16:bc:3d:aa:57)
    .... .... 0000 = Fragment number: 0
    0001 1011 1010 .... = Sequence number: 442
    [WLAN Flags: .p....F.]
    ▶ TKIP parameters
  ▼ Data (151 bytes)
    Data [...]: faf3c2183a608fb0204d9bfd9dd9a6e3d63d9ba5df30fc71d55e87a7617ce488579558d4537
    [Length: 151]
```

6.Conclusion

This exercise provided practical understanding of the WPA2 client connection process through Wireshark packet analysis. The capture was used to identify the major stages of wireless network discovery, authentication, association and WPA2 key establishment. Analysis of the RSN Information Element helped identify the advertised security configuration, while the EAPOL frames demonstrated the 4-way handshake used to establish the keys required for protected communication.